

Vayu VY-6

Architecture and System Design Description

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Companion to VAY-HF-0001, Flight Deck and Human Factors Design Description,
Issue 1

DEMONSTRATION ARTICLE. The Vayu VY-6 is a representative six-passenger electric vertical take-off and landing air taxi constructed by Safety Lab Aero. Aircraft-level characteristics follow published information for aircraft in this class and are listed against their sources in Appendix B. The system architecture below aircraft level is representative, and is not the design data of any manufacturer.

This document describes the design and stops where the assessment begins. Appendix A states what it therefore does not contain, and why each omission is deliberate.

Prepared by Safety Lab Aero.

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1 Aircraft Description and Concept

The Vayu VY-6 is a six-passenger electric vertical take-off and landing aircraft sized around a single operating idea: short revenue legs between elevated pads inside a city, flown by one pilot, many times a day. Every significant configuration choice follows from that sentence.

An aircraft that must hover out of a confined urban pad and then cruise on a wing cannot treat either regime as the exception, and an aircraft that flies eight to fourteen legs a day on stored energy cannot treat the energy store as a convenience. Those two consequences - two flight regimes with a transition between them on every sector, and a battery installation that is a primary structure and a primary heat source at once - shape the rest of the design.

The aircraft carries eight lift rotors on two longitudinal booms for the hover regime and two wing-mounted cruise propellers for wing-borne flight. The lift rotors are stopped and indexed in cruise; the cruise propellers are not turning in the hover. Nothing on the aircraft changes its geometry to make that happen: there is no tilting mechanism, and the transition is a matter of control law and airspeed rather than of moving structure.

Vayu VY-6 General Arrangement

Plan, profile and front elevation at one common scale. Dimensions per section 1.2.

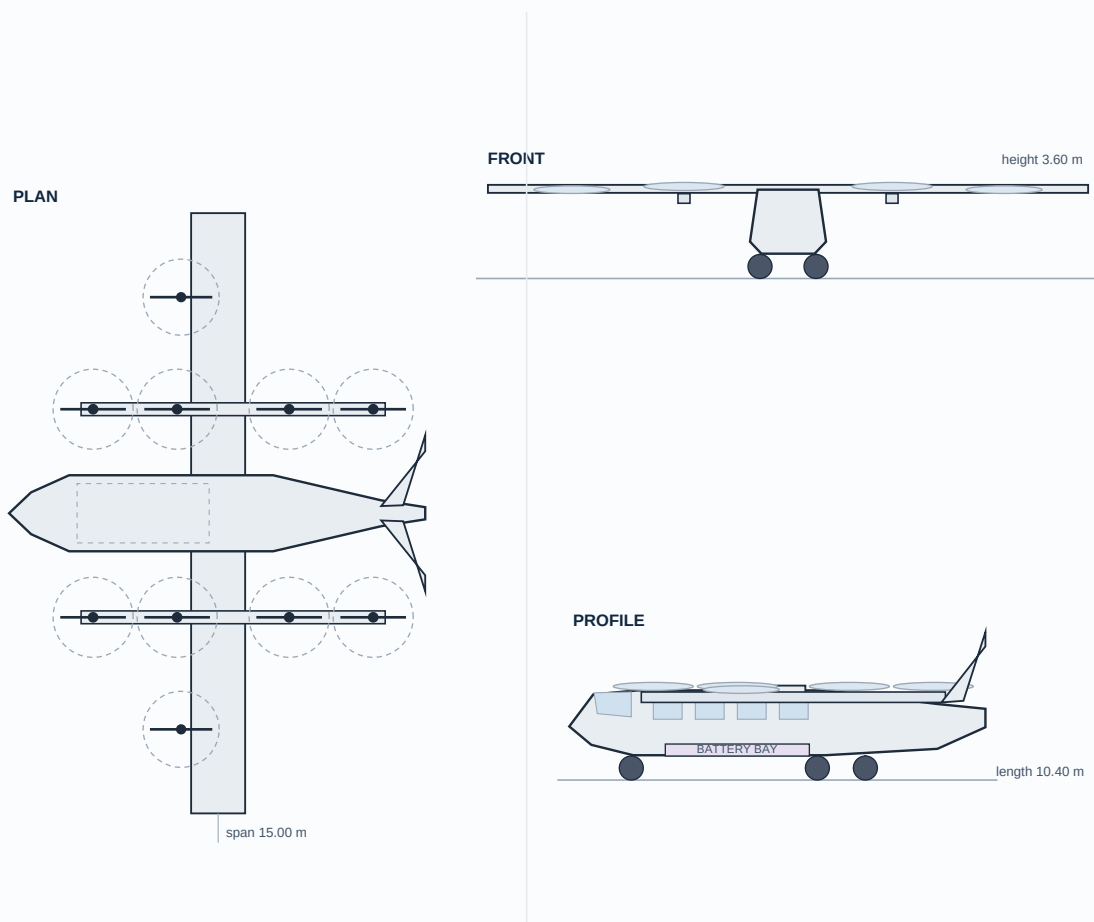


Figure 1-1 General arrangement. Plan, profile and front elevation to the dimensions of section 1.2, drawn at one common scale. There is no isometric.

1.1 Mission and Operating Concept

The aircraft flies point-to-point urban sectors of 20 to 30 km between elevated vertiport pads, typically eight to fourteen sectors per day, by day and by night. Cruise altitude is between 300 and 600 m above the city; the aircraft is not a high-altitude machine and does not need to be. A representative sector is nine minutes of wing-borne cruise between a two-minute vertical departure and a three-minute approach and landing.

Pads are elevated structures surrounded by buildings. The wind environment at the pad is shaped by those buildings rather than by the free-stream, and shear and turbulence off nearby structures are a normal operating condition rather than an exception. The aircraft is expected to depart and arrive into that environment on schedule.

Seven occupants are carried: one pilot on the flight deck and six passengers in the cabin. There is no cabin crew. Passengers board and leave on the pad with the aircraft powered, and the pilot remains in the seat throughout the turnaround.

Turnaround is about ten minutes at a pad with no line maintenance. The aircraft is expected to be self-sufficient between charges: it powers up, departs, flies the sector and returns on its own stored energy, and the operating concept assumes a charge is available only at defined points in the network rather than at every pad.

Range grows with battery generation. The airframe and its installations are sized for the longer-range store that is expected to follow, and the present energy allowance is a property of the fitted packs rather than of the aircraft.

1.2 Configuration and Principal Characteristics

Characteristic	Value	Characteristic	Value
Configuration	Lift plus cruise, fixed geometry	Wing span	15.00 m
Overall length	10.40 m	Overall height	3.60 m
Maximum take-off mass	2,600 kg	Payload	680 kg
Occupants	1 pilot, 6 passengers	Cabin crew	None
Lift propulsors	8 rotors, 2.00 m diameter	Cruise propulsors	2 propellers, 1.90 m diameter
Lift rotor arrangement	4 per boom, 2 booms	Cruise arrangement	Wing-mounted, one per side
Energy storage	2 independent packs, 4 strings each	Distribution	2 high-voltage sides, 1 tie
Flight control	Fly-by-wire, no mechanical reversion	Actuation	Electromechanical
Cruise speed	200 km/h	Maximum speed	250 km/h
Typical sector	20 to 30 km	Sectors per day	8 to 14
Landing gear	Retractable tricycle	Recovery	Ballistic parachute installed

Masses and dimensions are design values for the demonstration article. Aircraft-level characteristics follow the published envelope for aircraft of this class; Appendix B lists each and its source. No performance figure in this table is a guaranteed value and none is a safety objective.

1.3 Certification and Safety Basis

The intended certification basis is a special condition for small-category vertical take-off and landing aircraft, as applied by the certifying authority for the programme. The applicable category within that basis depends on the intended operation and on maximum passenger seating, and it is not settled in this document. The

aircraft is designed for passenger-carrying operations over a built-up area; what that implies for the category, and what the category implies for the objectives, is the safety assessment's to establish.

The safety assessment for the programme uses a five-class severity scale, the classes being No Safety Effect, Minor, Major, Hazardous and Catastrophic. This document names the scale so that the reader knows which vocabulary the assessment will use. It does not apply it. No failure condition is identified here, no severity is assigned, no probability objective is stated and no development assurance level is allocated.

The distinction matters for how this document should be read. Where the text says that two channels are independent, or that a bay is separately ventilated, it is stating a design fact that the assessment may rely on. Where the assessment needs a number - an exposure time, a crew response, a demonstrated rate - that number is not here, and the assessment records it as an assumption.

2 System Architecture Overview

2.1 System Inventory

The aircraft is divided into sixteen systems for design and description. The division is by responsibility rather than by physical location: a system may appear in several zones, and a zone commonly holds parts of several systems. Section 3 gives the zones and section 6 shows both together.

Each system carries a three-letter code. The codes are used throughout sections 5 and 6 and in the interface matrix below, and they are the identifiers by which the systems are referred to in every other programme document.

Code	System	Zones	Scope
DEP	Distributed Electric Propulsion	300, 400, 500, 600, 700	Inverters and motor control for the eight lift units and the two cruise units.
LFT	Lift Unit and Rotor Assembly	400, 500	Eight lift rotors, hubs, blade retention, parking brakes and blade indexing.
HVB	High Voltage Battery and Energy Storage	800	Two packs of four strings, battery management, enclosure, venting and charge interface.
EPD	Electrical Power Distribution	300, 800	High-voltage buses and tie, conversion to low voltage, essential bus, isolation monitoring.
TMS	Thermal Management	300, 400, 500, 800	Two coolant loops with a crossover serving batteries, inverters and motors.
FCS	Flight Control System	100, 300	Flight control computers, hover and wing-borne control laws, transition, command voting.
ACT	Flight Control Actuation	600, 700, 900	Electromechanical actuation of ruddervators, ailerons and flaps.
NAV	Navigation and Air Data	100, 300	Inertial and satellite position, air data, low-speed sensing, radar altimetry.
COM	Communications	100, 200, 300	Radios, vertiport datalink, audio panel, passenger address and cabin intercom.
DAA	Detect and Avoid	100, 300	Cooperative traffic, optical detection, terrain and obstacle data, surveillance processing.
DIS	Flight Deck Displays and Controls	100	Display surfaces, standby display, inceptors and the presentation of alerts.
LGS	Landing Gear and Ground Contact	1000	Fixed tricycle gear, wheel brakes, weight-on-wheels sensing and ground handling provisions.
STR	Airframe Structure	100 to 1000	Fuselage, wing box, booms, V-tail, battery bay structure and the load paths between them.
CAB	Cabin and Occupant Safety	200	Seats, restraints, doors and latch sensing, interior, stowage and the cabin environment.
ERS	Emergency Recovery System	300	Ballistic parachute, canister, harness, arming unit, inhibit logic and the pilot handle.
HMS	Health Monitoring and Data Recording	300	Continuous acquisition from both data buses, crash-protected recording and turnaround reporting.

2.2 System-to-System Interface Matrix

The matrix below lists the interfaces that cross a system boundary. It is not a wiring list: it names what is exchanged and in which direction, so that a reader can see which systems depend on which without reading both descriptions in full. Interfaces internal to one system are described in that system's subsection 5.x.4 and do not appear here.

From	To	What crosses the boundary
HVB	EPD	High-voltage supply from each pack to its own bus side.
EPD	DEP	High-voltage supply to the inverter groups on both bus sides.
EPD	ACT, NAV, DIS, COM	Low-voltage supply from the two low-voltage buses and the essential bus.
DEP	LFT	Motor drive to the eight lift units, and the stop-and-align command path.
FCS	DEP	Torque demand for all ten propulsors, and the received unit state.
FCS	ACT	Surface demand to the ruddervators, ailerons and flap drive.
NAV	FCS	Position, attitude, air data and low-speed state vector.
DAA	DIS	Traffic, terrain and obstacle picture for presentation.
LGS	FCS	Weight-on-wheels state used to select ground or air behaviour.
TMS	HVB, DEP	Coolant to the battery cold plates, the inverter plates and the motor jackets.
CAB	DIS	Door latch state for presentation on the flight deck.
LGS	ERS	Ground state to the inhibit logic.
FCS	DIS	Mode, regime and system state for presentation.
All systems	HMS	Parameter reporting on data bus A and data bus B.
STR	LFT, DEP, HVB	Mounting and load paths for the booms, the propulsors and the battery bay.

3 Zonal Breakdown

The aircraft is divided into ten zones. A zone is a physical volume with a boundary that something has to cross, and the boundaries are chosen so that they correspond to real separations in the structure rather than to convenient lines on a drawing.

The battery bay is treated as its own zone rather than as part of the cabin section it sits beneath, because its boundary is a real one: the bay is sealed from the cabin and vented overboard. The two booms are separate zones from each other and from the wing, because a boom is a distinct structure carrying its own installations along its own length.

Zone	Name	Contains	Boundary
100	Flight deck	Displays, inceptors, flight control computers, navigation and surveillance units.	Pressure-free bulkhead at the cabin front; open to the cabin above seat height.
200	Cabin	Six passenger seats, restraints, doors, interior linings and stowage.	Cabin floor above the battery bay; bulkhead to the aft equipment bay.
300	Aft equipment bay	Power distribution, thermal units, recorder, recovery canister.	Bulkhead at the cabin rear; skin panels with external access.
400	Left boom	Four lift units, boom harness, coolant branch, boom structure.	Boom attachment to the wing box; boom skin along its length.
500	Right boom	Four lift units, boom harness, coolant branch, boom structure.	Boom attachment to the wing box; boom skin along its length.
600	Left wing	Cruise unit, aileron and flap actuation, wing harness.	Wing root rib at the carry-through; wing skin.
700	Right wing	Cruise unit, aileron and flap actuation, wing harness.	Wing root rib at the carry-through; wing skin.
800	Underfloor battery bay	Two battery packs, battery management, cold plates, vent path.	Sealed from the cabin above; vented overboard through the lower skin.
900	V-tail	Ruddervator actuation and its harness.	Tail attachment frames; tail skin.
1000	Gear bays	Nose and main gear units, wheel brakes, weight-on-wheels sensing.	Bay walls and fairings; open to the airflow below.

4 Zone-Spanning Routings

Six routings cross zone boundaries. They are listed here because a routing that passes through several zones is the mechanism by which an event in one zone reaches equipment in another, and because the separation between routings is a design property rather than an accident of installation.

R-1 carries high-voltage supply from the battery bay to the two booms. It leaves zone 800, passes through the wing carry-through, and divides at the wing root into a left and a right run. The left run serves zone 400 and the right run serves zone 500; the two runs are on opposite sides of the carry-through and share no common conduit outboard of it.

R-2 carries high-voltage supply from the battery bay to the two wing-mounted cruise units. It follows the same carry-through as R-1 but is separately supported and separately sleeved within it.

R-3 is the low-voltage distribution. It runs from the aft equipment bay forward along both sides of the cabin to the flight deck, and outboard to the wing and the booms. The two sides of the low-voltage system run on opposite sides of the fuselage for the length of the cabin.

R-4 is the coolant routing. Two loops leave the aft equipment bay, one to the battery cold plates in zone 800 and one to the inverter plates and motor jackets in zones 400, 500, 600 and 700. The crossover valve that connects the two loops sits in zone 300.

R-5 and R-6 are data bus A and data bus B. Both reach every system. They run on opposite sides of the fuselage and take different paths through the wing carry-through, so that the two buses do not share a support, a penetration or a conduit anywhere along their length.

5 System Design Descriptions

The sixteen systems of section 2.1 are described below in the same order and to the same six headings, so that any two can be compared without re-reading both in full.

5.1 Distributed Electric Propulsion (DEP)

5.1.1 Purpose

DEP converts electrical power into torque and speed for the VY-6's ten propulsors: eight lift rotors on the left and right booms (zones 400, 500) and two wing-mounted cruise propellers (zones 600, 700), via ten inverter-motor units converting high-voltage DC to variable-frequency AC.

DEP turns torque and speed demands from FCS into per-motor current commands, returning motor and inverter status to FCS over the data buses. DEP holds no authority over flight path; it executes commands and reports propulsor state.

5.1.2 Description & Architecture

Each drive unit sits beside its propulsor: four per boom for lift rotors, one per wing root for cruise propellers. Each inverter takes high-voltage DC from EPD, produces three-phase AC, and drives a permanent-magnet motor coupled directly to the hub, no gearbox.

Motor position and current are sensed locally in an independent current-control loop per inverter. Each motor controller takes torque and speed setpoints over the data bus and reports winding temperature, DC link voltage, and fault flags.

5.1.3 Redundancy & Reconfiguration

Each of the ten propulsors has its own inverter-motor unit and current-control loop, so a fault confined to one unit affects only its propulsor. Lift units sit four to a boom; the remaining rotors on both booms operate independently of any one unit.

The two cruise propellers are fed from separate inverters and separate sections of the EPD high-voltage bus; a fault on one inverter does not reach the other propeller's drive path. FCS adjusts torque distribution across the remaining active propulsors.

5.1.4 Interfaces

DEP takes torque and speed commands from FCS over data buses A and B (routes R-5, R-6), one channel per boom and wing, returning motor and inverter status on the same buses. High-voltage power arrives from EPD on route R-1 to the booms and R-2 to the wings.

Coolant for inverter power stages and motor windings comes from TMS on route R-4, separate supply and return lines per installation. LFT interfaces at each rotor hub, taking drive torque from DEP and stop-and-align commands relayed through DEP.

5.1.5 Installation & Segregation

The four left-boom units occupy zone 400 and the four right-boom units zone 500, mounted outboard with cooling lines along the boom spar. The two cruise units sit at the wing root in zones 600 and 700.

Wiring and coolant for the left boom and wing are routed separately from the right boom and wing, per the aircraft's left/right split. Data bus A and bus B run on opposite sides of the fuselage centerline.

5.1.6 Operation

In hover, DEP drives all eight lift rotors at speeds and torques set by FCS, producing vertical thrust and pitch, roll and yaw control through differential thrust, while cruise propellers idle. During transition, DEP shifts torque from the lift rotors toward the cruise propellers as wing-borne lift builds.

In cruise, the lift rotors are stopped and aligned, drawing no torque, and the cruise propellers hold the thrust for the selected airspeed. DEP reports motor and inverter parameters to FCS throughout.

5.2 Lift Unit and Rotor Assembly (LFT)

5.2.1 Purpose

LFT comprises the eight lift rotors, hubs, blade retention hardware, and the stop-and-align mechanism fixing each rotor in a low-drag position for wing-borne flight. Four rotors mount on each boom, zones 400 and 500, each 2.0 meters in diameter, driven directly by a DEP motor.

The system turns motor torque into lift and control thrust in hover and transition, then stops rotation and aligns blades along the boom axis for cruise. LFT carries no drive electronics; torque comes from, and is monitored by, DEP.

5.2.2 Description & Architecture

Each rotor has two blades on a central hub via a retention pin and pitch bearing, mounted on its DEP motor's output shaft. A centrifugal blade-retention path independent of the pitch bearing means attachment does not rely solely on the bearing under rotation.

The stop-and-align mechanism is an electromechanical brake and index device in each hub, actuated by a signal relayed through DEP once FCS commands transition. It stops the rotor and locks blades aligned with local airflow, releasing when hover mode resumes.

5.2.3 Redundancy & Reconfiguration

The eight rotors form two sets of four, one per boom, fore and aft of the boom centerline. Each rotor's hub, blade retention hardware and stop-and-align mechanism share no moving parts with any other rotor's assembly.

A change in alignment status at one hub is confined to that rotor: the other seven retain independent stop-and-align mechanisms, each actuated by its own signal path from DEP. FCS recalculates achievable thrust and moment distribution across the remaining active rotors.

5.2.4 Interfaces

LFT takes drive torque directly from the DEP motor on each rotor, and stop-and-align actuation relayed through DEP from FCS. LFT returns rotor position, lock state, and blade condition through the associated DEP unit onto data bus A or B for that boom.

The hub and blade assemblies interface mechanically with the boom structure through the rotor mounting flange, sharing the boom's load path with the DEP motor housing. LFT has no direct connection to any other system.

5.2.5 Installation & Segregation

The four left-boom rotors sit in zone 400 and the four right-boom rotors in zone 500, each station separated by a structural bay that also carries the associated DEP unit. Blade sweep clearance is maintained between adjacent rotor stations on the same boom.

Left and right boom rotor assemblies are structurally and electrically independent, connected to the aircraft only through their boom root attachments and DEP data and power routes. No mechanical linkage exists between rotors on opposite booms.

5.2.6 Operation

During hover and transition, each rotor turns under torque from its DEP motor, blade pitch and speed set to produce the thrust and moments FCS commands. As transition proceeds, FCS commands each stop-and-align mechanism in sequence to stop the rotor and lock the blades along the boom.

In cruise, all eight rotors stay stopped and aligned, blades edge-on to reduce drag. On the return transition, FCS releases the mechanisms in sequence, and each DEP motor spins its rotor up before wing-borne lift is reduced.

5.3 High Voltage Battery and Energy Storage (HVB)

5.3.1 Purpose

HVB stores energy for propulsion, actuation, and aircraft systems, delivering high-voltage DC to EPD. Two independent packs sit in the underfloor bay, zone 800, each built from four parallel strings of series-connected cells in its own enclosure.

Each pack's battery management system monitors cell voltage, temperature and current, balances strings, and reports state of charge and health to EPD and FCS. Enclosure venting directs any gas released by a cell to a dedicated overboard path away from occupied zones.

5.3.2 Description & Architecture

The two packs sit side by side under the cabin floor, each with four strings joined in parallel at the pack's output contactors. Cells form prismatic modules, each wired for voltage and temperature sensing to the pack's controller.

Each pack's controller aggregates data from its four strings, computes state of charge and health, and commands the pack's main and precharge contactors. It communicates with EPD over the data buses, reporting status and receiving isolation commands.

5.3.3 Redundancy & Reconfiguration

The two packs are electrically independent from cell level through the output contactors, each with its own controller, cooling circuit, and vent path. Either pack may be isolated from the high-voltage bus at its own contactors without action at the other.

Within each pack, the four strings are individually fused and separately monitored, so an abnormal string is identified independently of the other three. EPD combines status from both packs to determine which bus sections remain fed.

5.3.4 Interfaces

HVB delivers high-voltage DC to EPD's main bus through the pack contactors, one per pack, and takes precharge and contactor commands from EPD over data bus A and B, one per pack. Battery data (charge, health, cell temperature) goes to EPD and on to FCS.

Coolant for the cell modules comes from TMS on separate supply and return lines per pack, on route R-4 from the aft equipment bay through zone 800. Each pack's vent path is independent, ducted to its own outlet.

5.3.5 Installation & Segregation

The left and right packs occupy separate bays in the underfloor compartment, zone 800, divided by a bulkhead that also carries each pack's vent ducting to its own outlet. Each pack's high-voltage cabling to EPD runs on its own side of the bulkhead.

Coolant lines, data bus connections, and vent ducting for the left and right packs stay on separate sides of the underfloor bay, matching the left/right segregation used for the booms and wings.

5.3.6 Operation

Normally both packs connect to the high-voltage bus and share the load from DEP, EPD's low-voltage conversion, and ACT, each controller balancing its own four strings continuously. Charge and health data are displayed to the crew and used by FCS for energy planning.

Between legs, ground charging equipment connects through EPD, and each pack accepts charge current independently under its own controller. Coolant flow to each pack is modulated by TMS from that pack's reported temperature.

5.4 Electrical Power Distribution (EPD)

5.4.1 Purpose

EPD distributes power from HVB to the DEP inverter-motor units, ACT actuators, and low-voltage systems including FCS and NAV. It comprises the high-voltage bus, its contactors, two low-voltage converters, and isolation monitoring.

EPD's contactor logic determines which bus sections are fed from which pack, and converts high-voltage power to low-voltage DC for route R-3. Isolation monitoring measures resistance between the high-voltage bus and airframe structure, reporting it to FCS.

5.4.2 Description & Architecture

The high-voltage bus has two sections, one fed primarily by the left pack and one by the right, joined by a bus-tie contactor. From each section, feeders extend to the boom lift inverters on route R-1 and to the wing cruise inverters and actuation loads on route R-2.

Two low-voltage converters, one per bus section, step high-voltage DC down for route R-3 to FCS, NAV, cabin systems and lighting. Isolation monitoring units measure leakage resistance on each section independently, reporting to FCS.

5.4.3 Redundancy & Reconfiguration

The two bus sections can be joined through the bus-tie contactor or separated, each then fed only by its own pack. Separating them confines a condition on one section's isolation monitoring to that section, leaving the other section's DEP and ACT feeders energized from its own pack.

Either low-voltage converter can supply the full low-voltage bus alone; normally both run in parallel, and each can be isolated at its own contactor. Feeder contactors to each boom and wing are independently controlled.

5.4.4 Interfaces

EPD takes high-voltage DC from HVB's two packs and issues contactor commands to HVB over data bus A and B. EPD supplies high-voltage power to DEP on routes R-1 and R-2, and low-voltage power to FCS, NAV, ACT electronics and cabin systems on route R-3.

EPD reports bus voltage, contactor state and isolation resistance to FCS over data bus A and B, and takes power configuration commands from FCS. TMS provides coolant to the converters and contactor housings on route R-4.

5.4.5 Installation & Segregation

The bus contactors and the two converters sit in the aft equipment bay, zone 300, with the bus-tie contactor central between the left and right feeders. Feeder cabling to each boom and wing exits the bay on the corresponding side.

Left-side and right-side feeders from the equipment bay to the booms and wings run along opposite sides of the fuselage structure, keeping the two bus sections separated along their length.

5.4.6 Operation

Normally the bus-tie contactor is closed, both packs and both converters share the aircraft's load, and isolation monitoring runs continuously on both sections. The crew views bus voltage, contactor state, and isolation resistance on flight deck displays.

During ground charging, EPD reconfigures contactors to route charge current from ground equipment to each pack under FCS supervision. In flight, EPD changes feeder and bus-tie states only on FCS command, based on flight phase and HVB/DEP status.

5.5 Thermal Management (TMS)

5.5.1 Purpose

TMS removes heat from the two battery packs, the ten DEP inverter-motor units, and the EPD low-voltage converters, rejecting it to ambient air through heat exchangers. Two independent coolant loops, pumps, heat exchangers, and route R-4 plumbing serve each cooled component.

Coolant temperature and flow are monitored per loop and reported to FCS, with pump speed modulated by the heat load reported by HVB, DEP and EPD. TMS maintains circulation whenever the corresponding electrical system is powered.

5.5.2 Description & Architecture

The two loops follow the aircraft's left/right split: one serves the left pack, the left boom's four DEP units, and the left converter; the other serves the corresponding right-side components, each loop with its own pump, heat exchanger, and reservoir, in zone 300.

Coolant is a single-phase liquid through cold plates bonded to the battery modules, inverter stages, and motor windings, returning through the heat exchanger where external airflow removes the heat. Sensors sit at each component's inlet and outlet.

5.5.3 Redundancy & Reconfiguration

The left and right loops are hydraulically separate from pump to heat exchanger, sharing no common plumbing, so flow in one loop is independent of conditions in the other. Each loop's pump and heat exchanger can be commanded off independently by that loop's controller.

Within a loop, the cold plates for the battery pack, the four DEP units, and the converter are plumbed in parallel branches from a common manifold, so one branch can be throttled or shut without affecting the others.

5.5.4 Interfaces

TMS supplies and returns coolant on route R-4 to HVB, DEP and EPD, with separate supply and return lines per component within a loop. TMS reports coolant temperature, flow rate and pump status to FCS over data bus A for the left loop and bus B for the right.

TMS takes heat load and temperature data from HVB's controllers, DEP's inverter and motor sensors, and EPD's converter sensors, using it to set pump speed per loop. Power for the pumps and controllers comes from EPD over route R-3.

5.5.5 Installation & Segregation

The two pumps, heat exchangers and reservoirs sit in the aft equipment bay, zone 300, on opposite sides of the bay per the left/right loop split. Coolant lines from each pump run forward and outboard to the underfloor battery bay and the corresponding boom and wing.

Left-loop plumbing to the left pack, boom and converter runs along the left side of the fuselage structure, separated from the right-loop plumbing serving the corresponding right-side components.

5.5.6 Operation

Once HVB, DEP and EPD are powered, both loops circulate continuously, pump speed varying with heat generated during hover, cruise and transition. Heat load rises in hover, when all eight lift rotors and inverters are active, shifting toward the cruise propeller inverters in wing-borne flight.

Between legs, the loops keep circulating during ground charging to remove heat from charge current. The crew monitors coolant temperature per loop on flight deck displays, alongside battery and inverter temperatures.

5.6 Flight Control System (FCS)

5.6.1 Purpose

FCS computes and issues the commands that fly the VY-6 in hover, wing-borne cruise, and the transition between them, turning pilot and autopilot inputs into commands for DEP and ACT. It comprises the flight control computers, control laws per flight mode, and transition logic.

FCS combines pilot inputs with attitude, position and air data from NAV to compute commands for DEP's propulsors and ACT's actuators. No mechanical linkage exists between the flight deck and the propulsors or surfaces; commands pass through FCS electrically.

5.6.2 Description & Architecture

The flight control computers sit in the flight deck avionics bay, zone 100, running separate control laws for hover mode, wing-borne mode, and transition. In hover, the law allocates thrust and moments across the lift rotors through DEP; in wing-borne mode, across the cruise propellers and ACT-driven surfaces.

The transition law blends the hover and wing-borne laws as airspeed builds or decays, ramping rotor and propeller torque with the stop-and-align sequence commanded to LFT through DEP, from air data and rotor state.

5.6.3 Redundancy & Reconfiguration

FCS issues commands over both data bus A and B, addressing the left boom, right boom, left wing and right wing channels separately, command and status paths staying independent through to DEP and ACT.

The control laws recompute thrust and moment allocation continuously from status reported by DEP, LFT, and ACT, redistributing torque and deflection across units reporting normal status. Each data bus carries command and status independent of the other.

5.6.4 Interfaces

FCS commands DEP's ten units with torque and speed setpoints over data bus A to the left boom and wing and bus B to the right boom and wing, receiving motor and inverter status in return. FCS commands ACT's actuators with surface setpoints and receives position and status over the same buses.

FCS receives attitude, position, airspeed, altitude and low-speed air data from NAV; bus voltage, contactor state and isolation resistance from EPD; and battery, inverter, converter and coolant status from HVB, DEP and TMS.

5.6.5 Installation & Segregation

The flight control computers occupy zone 100 alongside the pilot's controls and displays, with data bus A and B leaving the flight deck on opposite sides of the fuselage centerline toward the equipment bay, booms and wings.

Cabling for bus A follows the left side of the fuselage structure and bus B the right side, maintaining separation through to the boom, wing and equipment bay connection points.

5.6.6 Operation

In hover, FCS computes thrust and moment commands for the lift rotors from pilot stick and pedal inputs, using attitude and low-speed air data from NAV. In cruise, FCS computes surface commands for ACT and thrust commands for the cruise propellers, holding the selected airspeed and altitude.

During transition, FCS commands LFT's stop-and-align mechanisms through DEP once conditions permit, shifting authority from rotor thrust to aerodynamic surfaces. FCS presents propulsion, actuation, power, thermal and navigation status to the crew.

5.7 Flight Control Actuation (ACT)

5.7.1 Purpose

ACT positions the ailerons on the wings and the control surfaces on the V-tail in response to commands from FCS. Electromechanical actuators, one per surface, convert an electrical position command into surface deflection.

ACT reports achieved surface position and actuator status back to FCS, closing the position loop locally at each actuator while FCS sets the commanded position. The control surfaces have no mechanical or hydraulic linkage to the flight deck; deflection comes entirely from ACT.

5.7.2 Description & Architecture

Each aileron, left wing zone 600 and right wing zone 700, is driven by an actuator at the wing rib: a motor, gear reduction stage, and local position controller. The V-tail surfaces, zone 900, are each driven by a similarly configured actuator at the tail root.

Each actuator's controller closes a position loop from a sensor on the actuator output, taking its commanded position from FCS over the data bus. Wing and V-tail actuators share a common design, differing in gear ratio and output torque.

5.7.3 Redundancy & Reconfiguration

Each surface's actuator has its own motor, gear stage and position controller, independent of the others. The left and right aileron actuators are fed from separate feeders sourced from EPD's left and right bus sections, and commanded from separate data buses.

FCS recomputes command allocation from the position and status each actuator reports, adjusting deflection on surfaces reporting normal status. V-tail actuators are likewise fed from separate sources on the left and right sides of zone 900.

5.7.4 Interfaces

ACT takes surface position commands from FCS over data bus A for the left aileron and left V-tail surface, and bus B for the right aileron and right V-tail surface, returning position and status on the same buses. Power for the motors comes from EPD over routes R-2 and R-3.

TMS supplies no coolant to ACT directly; the actuators rely on surrounding airflow for heat rejection, with temperature reported to FCS. NAV's air data is used indirectly by FCS to schedule surface command gains at different airspeeds.

5.7.5 Installation & Segregation

The aileron actuators sit within the wing structure in zones 600 and 700, close to the surface driven, and the V-tail actuators sit at the tail root in zone 900. Left-side and right-side actuator wiring is routed separately back to the fuselage.

Bus A wiring to the left aileron and left V-tail actuators, and bus B wiring to the right aileron and right V-tail actuators, follow the same left/right split used elsewhere, keeping the two command channels separated.

5.7.6 Operation

In cruise and during transition, ACT positions the ailerons and V-tail surfaces continuously per FCS commands for roll, pitch and yaw control. In hover, aerodynamic surfaces contribute little authority, and ACT holds the positions FCS commands.

Actuator position and status are reported continuously to FCS in flight, and any change appears on the flight deck displays. Ground checks exercise each actuator through its travel range before departure, confirmed by FCS.

5.8 Navigation and Air Data (NAV)

5.8.1 Purpose

NAV determines the aircraft's position, attitude, velocity, and air data, and provides low-speed sensing for hover and transition when pitot-static air data is least reliable. Inertial sensors, satellite positioning, air data probes, and low-speed sensors feed FCS the state data needed for control commands.

NAV measures and reports aircraft state independently of the control laws that consume it; FCS uses that state to compute commands for DEP and ACT. NAV also feeds position and air data to crew displays and flight management functions.

5.8.2 Description & Architecture

An inertial measurement unit and satellite positioning receiver, both in zone 100, provide attitude, angular rate, position and velocity. Air data probes on the forward fuselage measure airspeed, altitude and angle of attack; low-speed sensors near the booms provide groundspeed and height.

The inertial and satellite positioning data combine in a navigation computer into a blended position and attitude solution, reported to FCS continuously. Air data probe and low-speed sensor outputs are processed separately, weighted by airspeed.

5.8.3 Redundancy & Reconfiguration

NAV has separate sensor sets for inertial/satellite positioning, forward air data, and low-speed sensing, each reporting to FCS independently. Low-speed sensors sit near the left and right booms, so groundspeed and height data stay available from either boom's sensor independently of the other.

FCS combines the navigation solution, air data, and low-speed sensor data by flight phase, weighting low-speed sensing more in hover and air data more in cruise. A change in one source's quality shifts the weighting applied.

5.8.4 Interfaces

NAV reports the blended navigation solution, forward air data, and low-speed sensor data to FCS over data bus A and B, sensor sets split per the aircraft's left/right convention. FCS uses this data for commands issued to DEP and ACT.

NAV takes low-voltage power from EPD over route R-3, with the inertial, satellite positioning, air data and low-speed sensor sets fed from separate points on the bus. NAV has no direct interface with HVB, DEP, LFT, ACT or TMS; its data reaches them through FCS.

5.8.5 Installation & Segregation

The inertial measurement unit, satellite positioning receiver and navigation computer sit in zone 100 alongside the flight control computers. Forward air data probes mount on the fuselage nose ahead of the flight deck, and low-speed sensors mount near the left and right boom roots in zones 400 and 500.

Wiring from the left-boom low-speed sensor to the navigation computer follows the bus A routing along the left side of the fuselage, and the right-boom sensor follows bus B along the right side.

5.8.6 Operation

Throughout flight, NAV continuously computes and reports position, attitude and velocity, low-speed sensors active during hover and transition and air data probes most relied upon in cruise. The crew views position, attitude, airspeed and altitude on flight deck displays fed by FCS.

During transition, FCS draws on both low-speed and forward air data as their relative reliability changes with airspeed. NAV's outputs continue through ground operations for taxi and the vertiport approach.

5.9 Communications (COM)

5.9.1 Purpose

The communications system links the flight deck to air traffic control, vertiport operations, and dispatch, and carries cabin address and crew intercom functions. It supports frequent short urban legs between vertiports, with routine calls to ground and tower and vertiport sequencing before each approach.

A vertiport datalink transceiver exchanges pad assignment, queue position, and charging bay status with the ground network automatically in range. The passenger address amplifier and cabin speakers carry boarding, taxi, and arrival announcements; the audio panel routes radio and intercom audio to the pilot headset.

5.9.2 Description & Architecture

Two VHF radios, a vertiport datalink transceiver, a crew audio panel, and a cabin address amplifier make up the system. The radios and transceiver mount in the aft equipment bay, zone 300, with antennas on the upper and lower fuselage skin in zone 100 and zone 200.

The audio panel sits in the zone 100 instrument panel, presenting radio and intercom selection through switches and a status readout. The cabin address amplifier drives speakers in the zone 200 sidewall trim; wiring runs from the equipment bay to the flight deck and cabin along route R-3.

5.9.3 Redundancy & Reconfiguration

The two VHF radios are wired to separate antennas and separate low-voltage feeds on route R-3, each selectable independently at the audio panel. The datalink transceiver operates on a third, dedicated antenna and feed, distinct from either voice radio channel.

Loss of one low-voltage feed removes only the equipment on that feed, since the radios and transceiver share no common feed segment upstream of the equipment bay distribution point. The audio panel draws from both feed segments through separate input diodes.

5.9.4 Interfaces

COM receives position and altitude data from NAV for datalink position reports, and traffic and terrain advisories from DAA the pilot can relay by voice. FCS supplies flight phase information used to sequence automatic cabin announcements without pilot input.

DIS presents COM channel selection and radio status on the flight deck displays. HMS records radio and datalink transmit and receive events, channel changes, and audio panel selections on data bus A. The amplifier also accepts a direct flight deck microphone input.

5.9.5 Installation & Segregation

The two radios, the datalink transceiver, and their harnesses are installed on opposite sides of the aft equipment bay, zone 300, separated by the bay's centerline structure. Antenna cables follow separate conduit paths to the upper and lower fuselage antennas, crossing zone 100 and zone 200.

Low-voltage feed wiring for COM runs along route R-3, kept apart from route R-1 and route R-2 high-voltage runs by standoff brackets at each frame station. Cabin address speaker wiring in zone 200 is routed clear of the door mechanism wiring.

5.9.6 Operation

Before departure the pilot selects the active radio and datalink channel at the audio panel and confirms cabin address function with a test announcement. During each leg the pilot monitors the selected radio; the transceiver exchanges pad and queue status with the destination vertiport automatically.

The amplifier plays a recorded boarding announcement at the gate and an arrival announcement after touchdown, with the pilot able to override either from the flight deck microphone. Between legs the pilot reviews radio and datalink status before requesting the next pad assignment.

5.10 Detect and Avoid (DAA)

5.10.1 Purpose

The detect and avoid system gives the pilot awareness of other air traffic, terrain, and fixed obstacles in the urban environment, where buildings, cranes, and rotorcraft can be close to the route. It combines a transponder-based traffic receiver with short-range sensors around the aircraft.

Traffic and obstacle information is fused into tracks passed to FCS for display formatting and to DIS for presentation on the flight deck screens. The system operates continuously from engine start to shutdown across the daily leg count.

5.10.2 Description & Architecture

DAA combines an ADS-B in/out transponder, short-range obstacle sensors around the airframe, and a track processor in the aft equipment bay, zone 300. Camera and lidar sensors mount at the nose, each wingtip, and the tail cone, giving overlapping coverage near pads.

The track processor correlates transponder returns, received directly and relayed through the COM datalink, with the local sensor picture to build a combined track list. This list refreshes continuously and passes over data bus A to the displays and to FCS.

5.10.3 Redundancy & Reconfiguration

The nose, wingtip, and tail sensors overlap in coverage, so the volume around the aircraft is observed by more than one location. The two wingtip sensor sets are wired to opposite low-voltage feed segments on route R-3, each on a separate input channel.

The track processor has two independent processing channels, each capable of building a track list from the full sensor set, with outputs cross-checked before reaching the displays. If the transponder path is affected, the processor builds tracks from local sensors alone.

5.10.4 Interfaces

DAA sends traffic and obstacle tracks to FCS for correlation with the planned flight path and to DIS, rendered on the traffic and terrain page the pilot monitors during pad approach and departure. NAV supplies aircraft position and attitude used to place tracks.

COM relays vertiport-sourced traffic reports to the DAA processor over the datalink when direct transponder reception is limited. HMS records the track list, sensor source composition, and processing channel changes on data bus A.

5.10.5 Installation & Segregation

The track processor is installed in the aft equipment bay, zone 300, on an independent shelf with its own low-voltage feed. The nose sensor sits in zone 100, wingtip sensors in zone 600 and zone 700, and the tail sensor in zone 900.

Sensor data cables run along route R-5, the data bus A path, kept separate from route R-1 and route R-2 high-voltage runs by standoff brackets. Wingtip sensor feed wiring follows the same left and right segregation used for wing high-voltage distribution.

5.10.6 Operation

During taxi and hover near a vertiport pad, the pilot monitors the traffic and terrain display for nearby structures, ground vehicles, and other aircraft, using track symbology to judge clearance. In cruise, the

display shows tracks relevant to the planned route and altitude.

The pilot can select expanded or compact display ranges, toggle terrain shading, and query a track for altitude and closure information. No pilot action is required to keep the track list current from engine start through shutdown.

5.11 Flight Deck Displays and Controls (DIS)

5.11.1 Purpose

The flight deck displays and controls system presents flight, navigation, propulsion, and system status to the pilot and provides the inceptors and switches used to command the aircraft. It brings together information from FCS, NAV, DAA, and the propulsion and power systems onto two display surfaces.

Primary and secondary flight information, system synoptic pages, and alert messages are shown across the displays, while inceptors and hard switches let the pilot command flight path, thrust, and system configuration. The system operates from before engine start through shutdown on every leg.

5.11.2 Description & Architecture

Two display surfaces mount side by side in the zone 100 instrument panel, each driven by its own graphics processor and configurable to flight, navigation, or synoptic pages through bezel keys. A center inceptor and throttle levers send electrical flight path and thrust commands to FCS.

An alerting panel above the displays presents master caution and warning lights and annunciator legends, backed by an aural tone generator sharing the COM audio panel path. Hard switches for gear, lighting, and system resets sit along the lower instrument panel.

5.11.3 Redundancy & Reconfiguration

The two displays are driven from separate graphics processors on separate low-voltage feeds from route R-3, and each processor can source content from either FCS computer independently. Either display can show any available page, including the other surface's page.

The center inceptor carries two independent signal paths to the two FCS computers through separate connectors; the throttle levers likewise send independent position signals to each computer. The alerting panel's tone generator and light drivers share the feed with the left display processor, the right processor on the other segment.

5.11.4 Interfaces

DIS presents what FCS reports for flight path, attitude, and system status, and renders traffic and terrain tracks from DAA on a dedicated page. NAV data reaches the displays through FCS rather than a direct connection.

COM channel and datalink status appear on a system synoptic page, and LGS gear position is shown on the same page during extension and retraction. HMS records display page selections, alert messages, and inceptor and throttle positions on data bus A and data bus B.

5.11.5 Installation & Segregation

The two displays, their processors, and the alerting panel install in the zone 100 instrument panel, with the left processor and display on one low-voltage feed segment and the right pair on the other, on separate bundles to the equipment bay junction.

The center inceptor and throttle wiring runs separate from display and processor wiring, following route R-5 and route R-6 for its two independent FCS signal paths. Hard switch wiring for gear, lighting, and resets connects to route R-3.

5.11.6 Operation

At power-up the pilot brings up both displays through a built-in test, then selects the primary flight page on one surface and a synoptic page on the other for taxi and takeoff. In cruise the pilot commonly leaves one display on navigation.

Alert messages appear on whichever display shows a synoptic or status page, with an aural tone and, for more significant messages, a caution or warning light. The pilot acknowledges alerts through a glare shield switch and can call up the associated system page.

5.12 Landing Gear and Ground Contact (LGS)

5.12.1 Purpose

The landing gear and ground contact system supports the aircraft on the ground and vertiport pads, absorbs touchdown at the end of each hover descent, and provides signals used elsewhere to recognize weight on the wheels.

A retractable, wheeled, four-point gear arrangement is used, with nose and main legs stowed into gear bays after takeoff and extended ahead of each pad approach. A manual tow fitting and parking brake support ground movement between pads.

5.12.2 Description & Architecture

Each of the three gear legs carries a wheel, a shock strut, and an electric retraction actuator supplied by ACT, stowing into an individual bay in zone 1000 after takeoff. The nose leg retracts forward and the two main legs retract inboard.

A weight-on-wheels sensor at each main leg reports ground or air state to FCS, and a position sensor on each actuator reports gear position to DIS. The parking brake acts on the main wheels through an electric caliper; a tow fitting on the nose leg allows repositioning without power.

5.12.3 Redundancy & Reconfiguration

The four legs are actuated by individually powered retraction motors, each fed from route R-3 through its own circuit, so one leg's cycling does not depend on another's circuit. Weight-on-wheels sensing is taken from both main legs independently.

Each leg carries a mechanical downlock engaged by spring force once the actuator reaches the extended position, independent of continued power to the actuator. A manual extension handle in the flight deck can drive all four legs to the downlocked position without the electric actuators.

5.12.4 Interfaces

LGS reports gear position and weight-on-wheels state to FCS, used in flight control logic, and to DIS for the gear indication. ACT provides the retraction and extension actuator function itself.

STR carries the gear leg attachment fittings and touchdown loads into the airframe primary structure at each of the four bay locations. HMS records gear extension and retraction events, weight-on-wheels transitions, and downlock indications on data bus B.

5.12.5 Installation & Segregation

The four gear bays occupy zone 1000, the nose bay forward of the flight deck and the two main bays beneath the wing root on each side, each bounded by its own structural frame.

Retraction actuator power wiring runs along route R-3 in individual circuits per leg to the main low-voltage distribution panel. Position and weight-on-wheels sensor wiring follows route R-5 to FCS and DIS, separate from actuator power within each bay.

5.12.6 Operation

The pilot commands gear retraction after a positive rate of climb following takeoff, and extension during approach to each pad, observing gear position on the flight deck display as each leg cycles.

On the ground the pilot sets the parking brake before shutdown; if repositioning is needed without power, ground crew connect the tow fitting and release the brake. The manual extension handle is used only if the electric actuators do not respond.

5.13 Airframe Structure (STR)

5.13.1 Purpose

The airframe structure carries the fuselage, wing, booms, V-tail, and battery bay, providing load paths connecting the lift rotors, cruise propellers, gear, and battery packs to a single airframe. It is built primarily from carbon fiber composite.

The structure supports the aircraft through the full range of flight conditions on the urban route network, from hover through transition to wing-borne cruise and back. Access panels throughout allow inspection and maintenance of systems within each zone.

5.13.2 Description & Architecture

The fuselage is a semi-monocoque composite shell from the flight deck, zone 100, through the cabin, zone 200, to the aft equipment bay, zone 300, with the underfloor battery bay, zone 800, between the main wing spars.

The wing consists of left and right panels, zone 600 and zone 700, joined at a center section through the fuselage above the battery bay, with cruise propellers on pylons ahead of the leading edge. A V-tail, zone 900, provides aft control surface attachment points used by ACT.

5.13.3 Redundancy & Reconfiguration

The wing center section carries bending and shear loads through upper and lower composite spar caps continuing across the fuselage, so a load can transfer through either spar depending on chord location. Boom attachment fittings at each wing panel are duplicated fore and aft.

The battery bay structure separates the two independent HVB packs with a composite bulkhead the length of the bay, each pack in its own structural bay beneath the cabin floor. Gear bay structure at each of the four locations is an independent frame.

5.13.4 Interfaces

STR provides mounting structure and load paths for LFT at each boom station, for DEP at the wing-mounted cruise propeller pylons, and for LGS at each of the four gear bays. HVB packs are secured within the zone 800 bay structure.

ACT control surface actuators mount to fittings on the wing trailing edge and V-tail structure; EPD high-voltage cabling for route R-1 and route R-2 passes through conduits in the boom and wing spar. HMS records structural strain gauge readings on data bus A.

5.13.5 Installation & Segregation

Structural bays and zones are separated by bulkheads and frames that also serve as segregation boundaries; the battery bay bulkhead separating the two HVB packs also carries route R-4 coolant lines on opposite sides.

Wing structure carries route R-2 high-voltage cabling to the cruise propeller pylons along the leading edge box, separated from the wing accessory bay by the forward spar. Access panels allow inspection without disturbing load-carrying members.

5.13.6 Operation

The structure is passive in normal operation, requiring no pilot action; ground crew assess its condition through visual inspection at access panels between legs. Strain sensors report continuously to HMS, adding to the turnaround readiness data.

Ground crew check gear bay, battery bay, and boom access panels during turnaround inspection for damage or loose fasteners before release. Scheduled structural inspections at longer intervals examine the primary spar and fuselage joints.

5.14 Cabin and Occupant Safety (CAB)

5.14.1 Purpose

The cabin and occupant safety system covers passenger seating, restraints, doors, and interior of the six-passenger cabin, along with the environmental controls that keep the space comfortable across the day and night schedule.

Six passenger seats are arranged in the cabin, zone 200, each with its own restraint, and two doors provide entry and exit at each vertiport stop. Interior trim, stowage, and cabin lighting complete the environment, along with address and audio functions provided by COM.

5.14.2 Description & Architecture

The cabin holds six forward-facing seats in a two-two-two arrangement, each with a four-point restraint and single-point buckle release. Two doors, one each side, open outward on hinges near the leading edge, held by an electric latch with a manual release lever.

Cabin lighting is LED strips along sidewall and ceiling trim, supplied from route R-3; environmental control outlets at the forward and aft cabin ends deliver conditioned air managed by TMS. Stowage compartments beneath each seat and in the aft cabin wall hold personal items.

5.14.3 Redundancy & Reconfiguration

Each of the two doors has an independent electric latch actuator and its own manual release lever, entirely separate from the other door's mechanism and wiring. Each seat restraint buckle and webbing set is an independent, replaceable unit.

Cabin lighting is arranged in two strings, one per sidewall, wired to opposite low-voltage feed segments on route R-3, with ceiling trim lighting providing overlap. The forward and aft environmental outlets are supplied by separate ducting runs from the TMS manifold.

5.14.4 Interfaces

CAB receives conditioned air from TMS at the forward and aft cabin outlets and low-voltage power from EPD for lighting and door latch actuators. COM provides the passenger address function through CAB sidewall speakers and the crew intercom link.

DIS presents door and latch status to the pilot on a cabin synoptic page during boarding and before departure. HMS records door open and closed events, latch status, and restraint buckle status where sensed, on data bus B.

5.14.5 Installation & Segregation

Cabin seats mount on tracks in the cabin floor structure, zone 200, above the underfloor battery bay, zone 800, with seat track fasteners separate from the battery bay mounting rails below. Door hinges and latches mount in the fuselage door frame structure.

Cabin lighting and door latch wiring runs along route R-3 within the sidewall trim, clear of the high-voltage route R-1 and route R-2 conduits beneath the cabin floor. Cabin address speaker wiring shares the trim raceway on a separate connector run.

5.14.6 Operation

Passengers board through either door at the vertiport pad; cabin crew are not carried, so the pilot or ground staff confirm each occupant is seated and restrained before requesting doors closed.

During flight the environmental outlets and cabin lighting operate continuously, adjustable by the pilot from the flight deck. On arrival the pilot confirms the aircraft is at the pad before releasing the door latches, and passengers exit through either door.

5.15 Emergency Recovery System (ERS)

5.15.1 Purpose

The emergency recovery system provides a ballistic parachute intended to lower the aircraft to the ground, with an activation handle in the flight deck, a control unit evaluating arming conditions, and a parachute assembly in the aft equipment bay, zone 300.

Arming and inhibit logic in the control unit uses airspeed, altitude, and attitude data from NAV and FCS to determine whether an activation command proceeds to the pyrotechnic charge. The deployment path runs through a frangible panel in the upper aft fuselage skin.

5.15.2 Description & Architecture

The parachute assembly consists of a folded canopy and riser lines in a canister, a pyrotechnic charge that propels a drogue from the canister, and a bridle to attachment fittings on the aft fuselage. The canister mounts in zone 300 beneath a frangible cover panel.

A control unit next to the canister receives an activation signal from a guarded handle at the flight deck center console and evaluates arming conditions before sending a firing signal. The path includes a separate safe and arm switch that must be armed, set by the pilot, before firing can reach the charge.

5.15.3 Redundancy & Reconfiguration

The activation handle and the safe and arm switch are two separate, independently operated flight deck controls, both required before a firing signal reaches the charge. The control unit receives airspeed and altitude from NAV over an input separate from the general data bus.

The firing circuit is powered from a dedicated battery pack within the control unit, separate from the aircraft's main low-voltage distribution on route R-3. Wiring from the unit to the charge runs in a shielded, dedicated harness.

5.15.4 Interfaces

ERS receives airspeed and altitude data from NAV and attitude data from FCS for the control unit's arming and inhibit logic. EPD powers the unit's monitoring functions; the firing circuit draws from the dedicated internal battery pack.

DIS presents control unit status, including armed state and fault indication, on a dedicated flight deck annunciator near the activation handle. HMS records handle guard state, safe and arm switch position, and control unit status continuously on data bus B.

5.15.5 Installation & Segregation

The canister, control unit, and dedicated firing battery install together in the aft equipment bay, zone 300, on a mounting structure attached to the fuselage frames beneath the frangible cover panel, apart from the COM and DAA equipment.

The firing harness between the control unit and the charge runs in a dedicated conduit separate from route R-3, route R-5, and route R-6. Activation handle wiring runs forward from zone 300 to zone 100, kept apart from general flight deck switch wiring.

5.15.6 Operation

The safe and arm switch is set by the pilot according to operator procedures for the phase of flight, and the guarded activation handle is available at the flight deck center console throughout.

Should the pilot lift the guard and pull the handle, the control unit checks current arming conditions before passing a firing signal to the charge, which deploys the drogue and canopy through the frangible panel in the aft fuselage skin. Control unit status is visible on the annunciator throughout.

5.16 Health Monitoring and Data Recording (HMS)

5.16.1 Purpose

The health monitoring and data recording system continuously gathers status and event data from every other system, records it for later review, and compiles a turnaround readiness report used by ground crew and the pilot between legs.

Data is collected from both aircraft data buses, route R-5 data bus A and route R-6 data bus B, and written to two independent recorders in the aft equipment bay, zone 300.

5.16.2 Description & Architecture

A data concentrator in zone 300 collects messages from data bus A and data bus B, timestamps them against a common clock, and writes them to two recorder units mounted alongside it. Each recorder stores a rolling record, retrievable through a data port.

The turnaround readiness report is generated at engine shutdown, drawing on status messages received during the preceding flight from every reporting system, and presented to ground crew on a maintenance terminal. The report lists system status entries by zone and system code.

5.16.3 Redundancy & Reconfiguration

The two recorders each connect to both data bus A and data bus B independently, so either can build a complete record without depending on the other. Each recorder has its own low-voltage feed from route R-3, on separate feed segments.

The data concentrator has two processing channels, each capable of building the turnaround readiness report from the collected bus data. If one data bus is affected, the concentrator builds its report from the remaining bus, and indicates which bus contributed each entry.

5.16.4 Interfaces

HMS records from every system on data bus B, and also data bus A, gathering status and event messages from DEP, LFT, HVB, EPD, TMS, FCS, ACT, NAV, COM, DAA, DIS, LGS, STR, CAB, and ERS.

DIS can call up recent recorded events for display on a flight deck maintenance page, drawing on the same recorder data used for the turnaround readiness report. EPD supplies low-voltage power to the concentrator and both recorders from route R-3.

5.16.5 Installation & Segregation

The data concentrator and both recorders install together in the aft equipment bay, zone 300, on a shelf separate from the COM, DAA, and ERS equipment. Each recorder connects to the bay's data bus A and data bus B points by an individual cable pair.

Recorder power wiring follows route R-3 in two separate segments back to the low-voltage distribution panel, apart from the data bus cabling on route R-5 and route R-6. The maintenance data port is accessible from an external panel near the equipment bay.

5.16.6 Operation

The concentrator and recorders operate automatically from engine start to shutdown, requiring no pilot action beyond calling up recent recorded events on the maintenance page if needed. At shutdown the concentrator compiles the turnaround readiness report.

Ground crew connect a maintenance terminal to the external data port between legs to review the turnaround readiness report before releasing the aircraft for its next flight, and retrieve recorder data at scheduled maintenance intervals.

6 Zonal Model and System Schematics

This section shows where the systems of section 5 are installed and how their units connect. Figure 6-1 places the ten zones of section 3 on the aircraft and draws the six zone-spanning routings of section 4 over the profile. Figures 6-2 to 6-17 give one block schematic per system, in the order of the inventory in section 2.1.

The schematics are architecture drawings rather than wiring diagrams. A box is a unit or a resource, a line is something that crosses between them, and the line type says what. Where a box is drawn in grey it belongs to another system and is shown only because the interface is the point of the figure.

Read together with section 5, the figures answer a question the text alone answers slowly: what does this system share with that one, and where does the sharing physically happen.

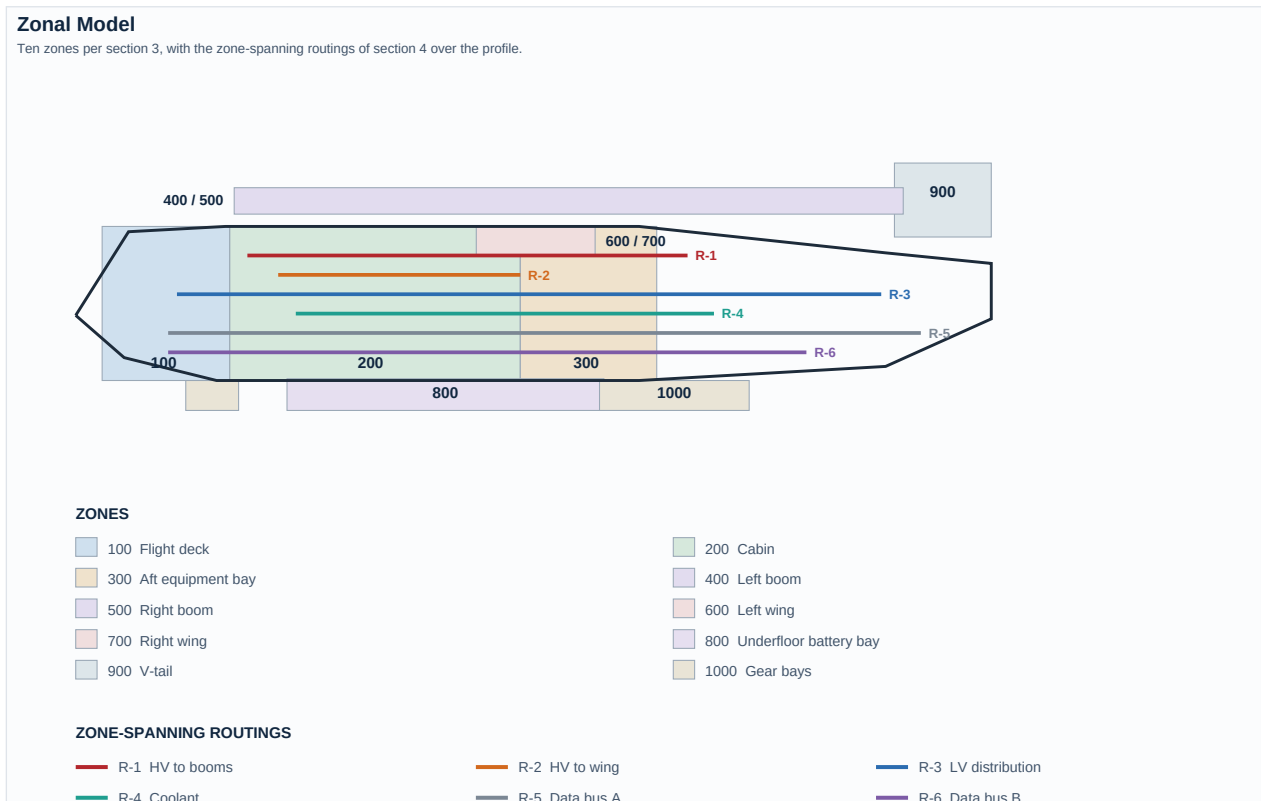


Figure 6-1 Zonal model. Ten zones per section 3, with the zone-spanning routings of section 4 drawn over the profile.

6.1 Distributed Electric Propulsion (DEP)

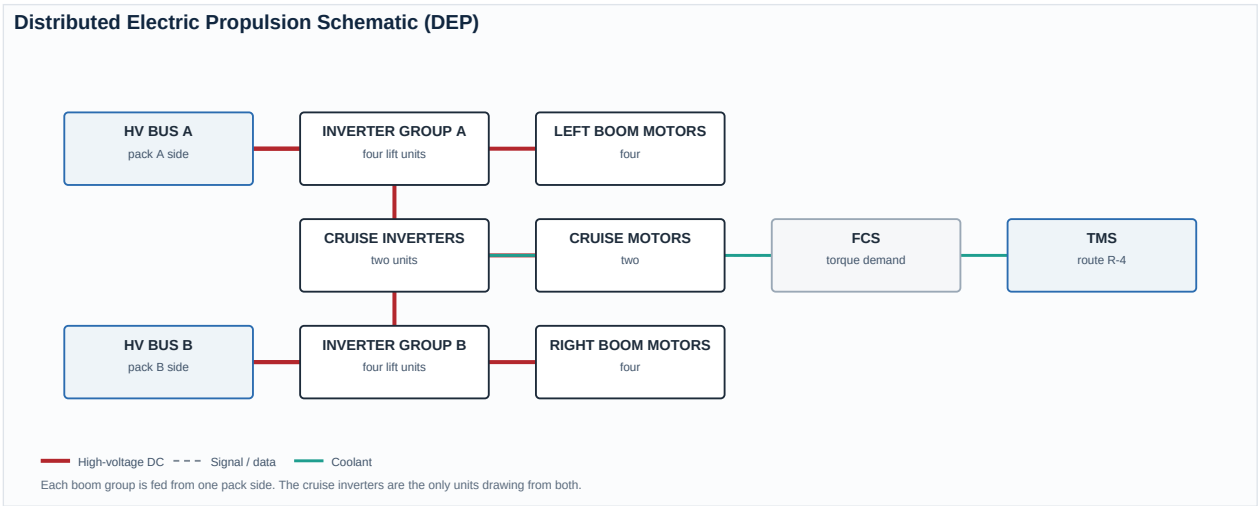


Figure 6-2 Distributed Electric Propulsion (DEP). Bus side to inverter group to motors, with the cruise inverters drawing from both sides.

6.2 Lift Unit and Rotor Assembly (LFT)

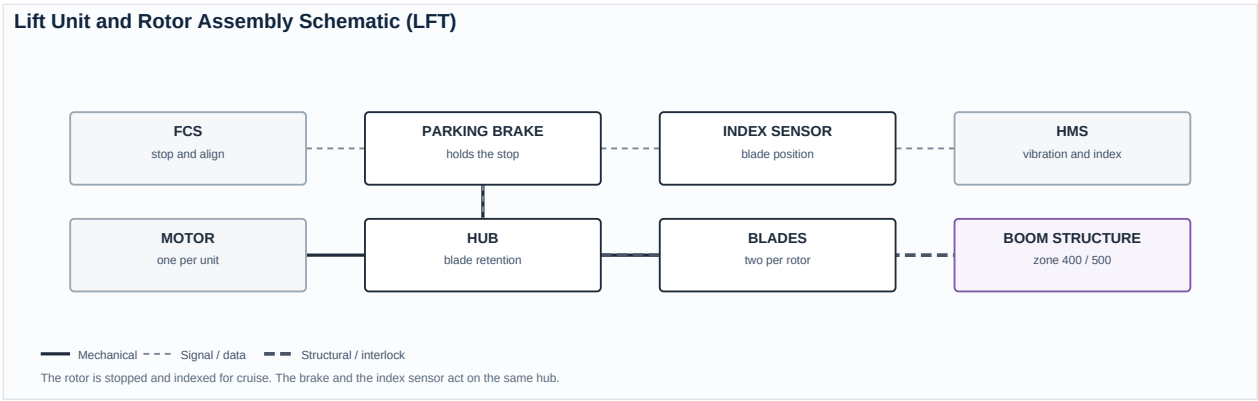


Figure 6-3 Lift Unit and Rotor Assembly (LFT). One lift unit: motor, hub, blades, parking brake and index sensor on the boom structure.

6.3 High Voltage Battery and Energy Storage (HVB)

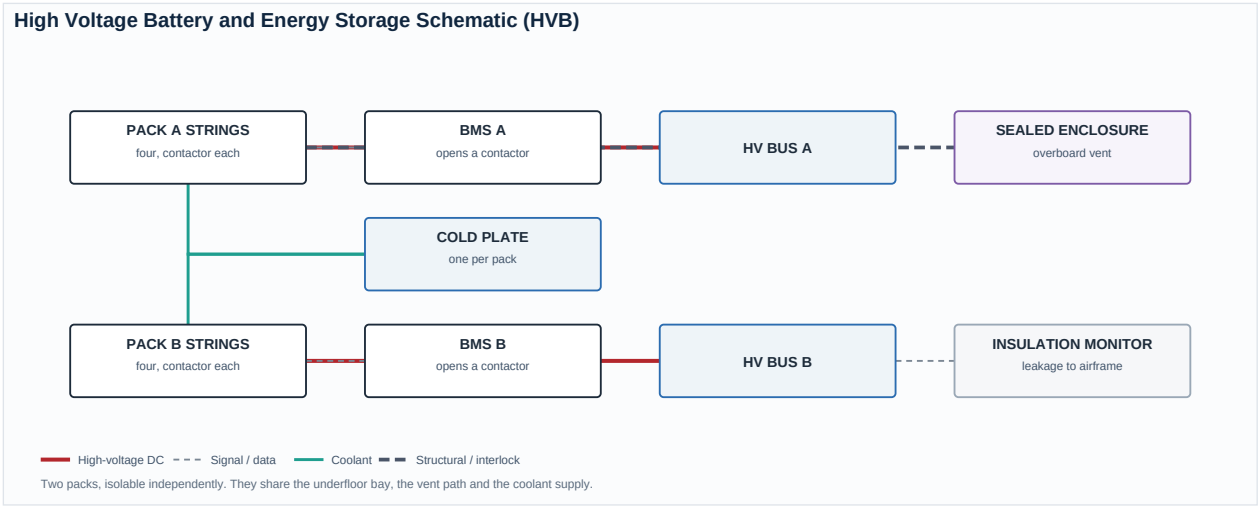


Figure 6-4 High Voltage Battery and Energy Storage (HVB). Two packs, their management units, the shared bay and the shared vent path.

6.4 Electrical Power Distribution (EPD)

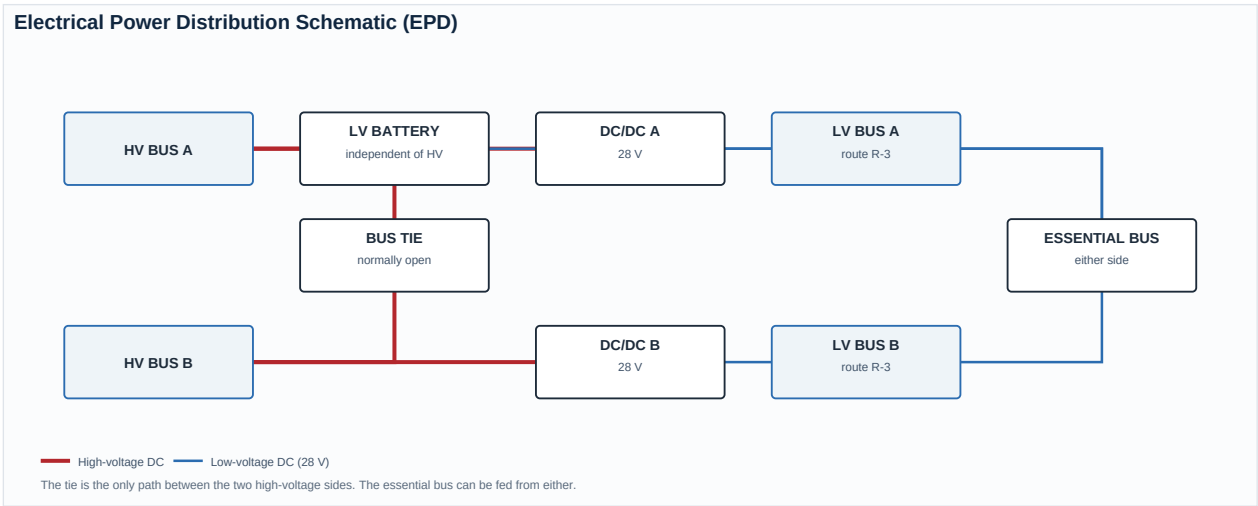


Figure 6-5 Electrical Power Distribution (EPD). Two high-voltage sides, the tie between them, conversion to low voltage and the essential bus.

6.5 Thermal Management (TMS)

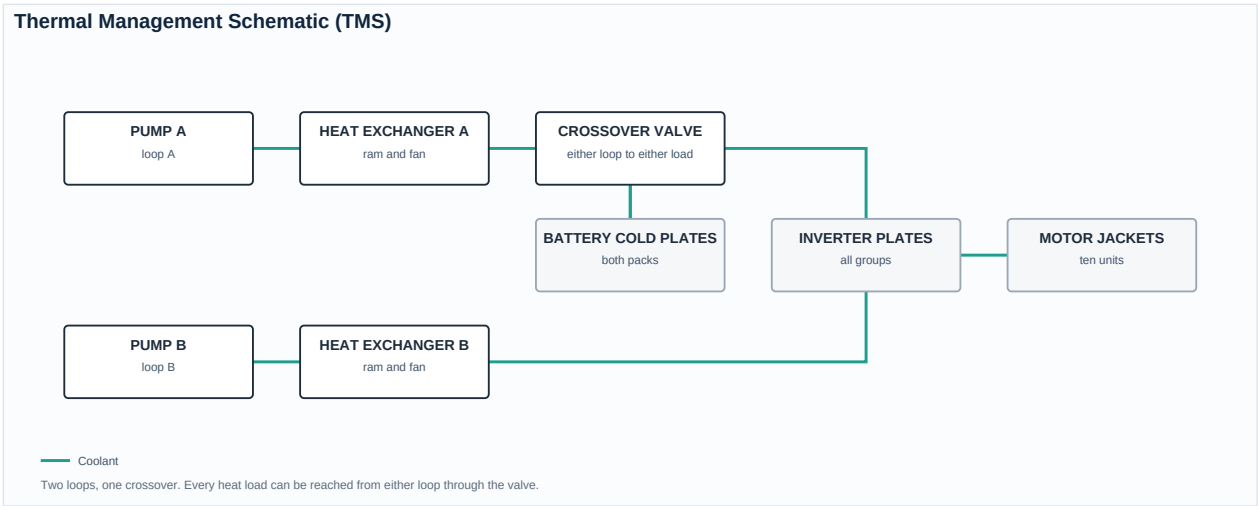


Figure 6-6 Thermal Management (TMS). Two coolant loops and the crossover valve that lets either loop reach either heat load.

6.6 Flight Control System (FCS)

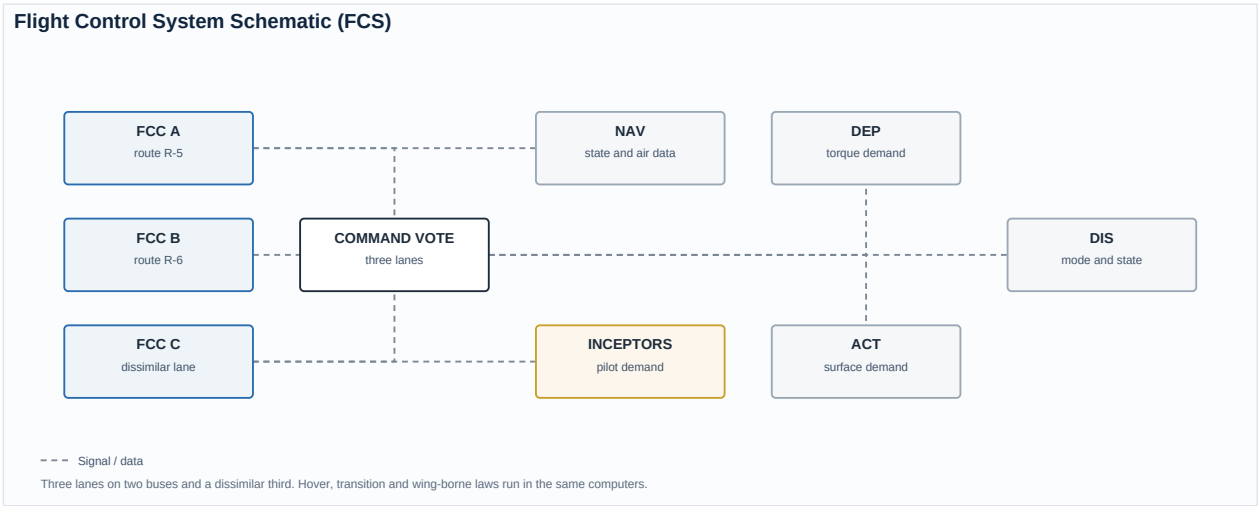


Figure 6-7 Flight Control System (FCS). Three computing lanes, the command vote, and what the vote drives.

6.7 Flight Control Actuation (ACT)

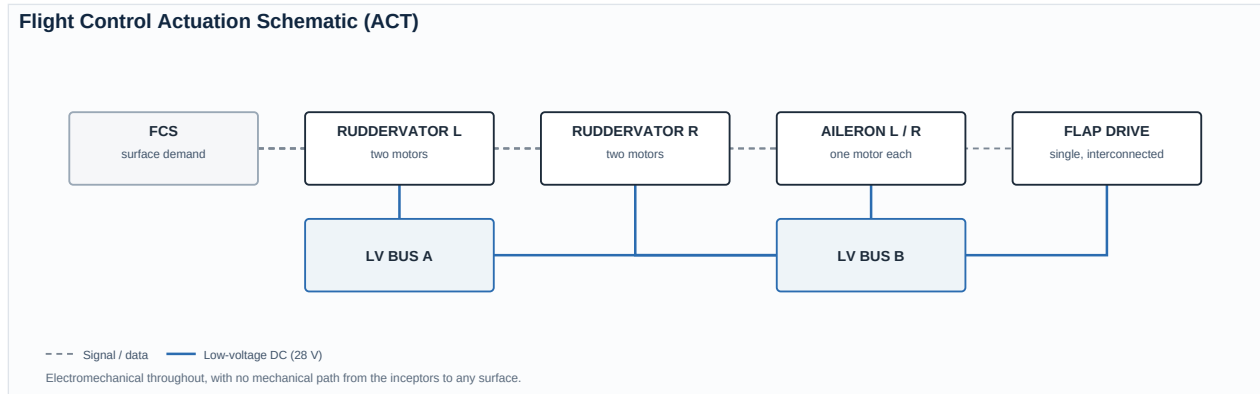


Figure 6-8 Flight Control Actuation (ACT). Surface actuation and the low-voltage side each actuator is fed from.

6.8 Navigation and Air Data (NAV)

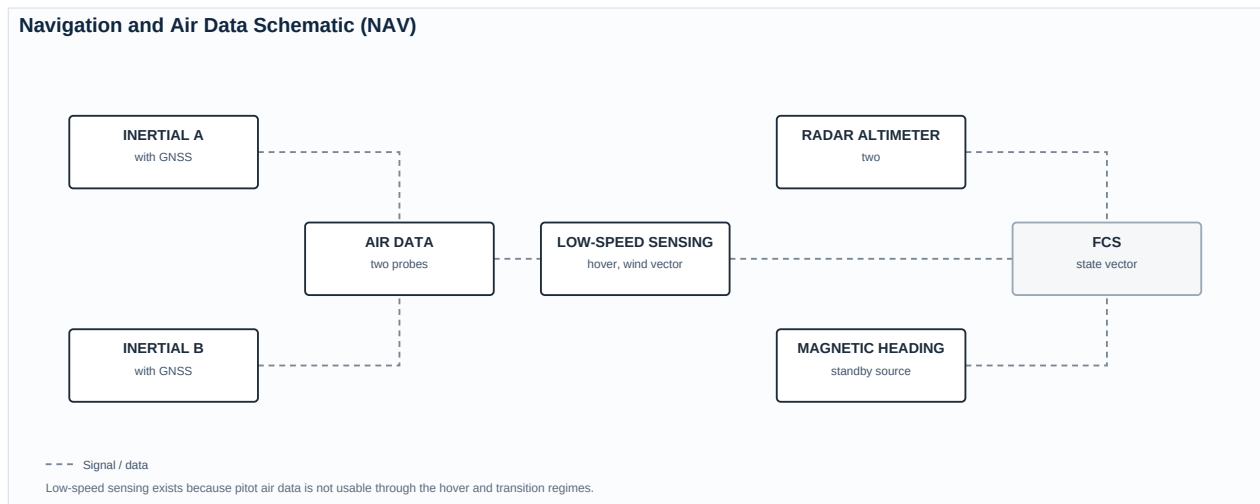


Figure 6-9 Navigation and Air Data (NAV). Inertial and air data sources, low-speed sensing, and what reaches the flight control computers.

6.9 Communications (COM)

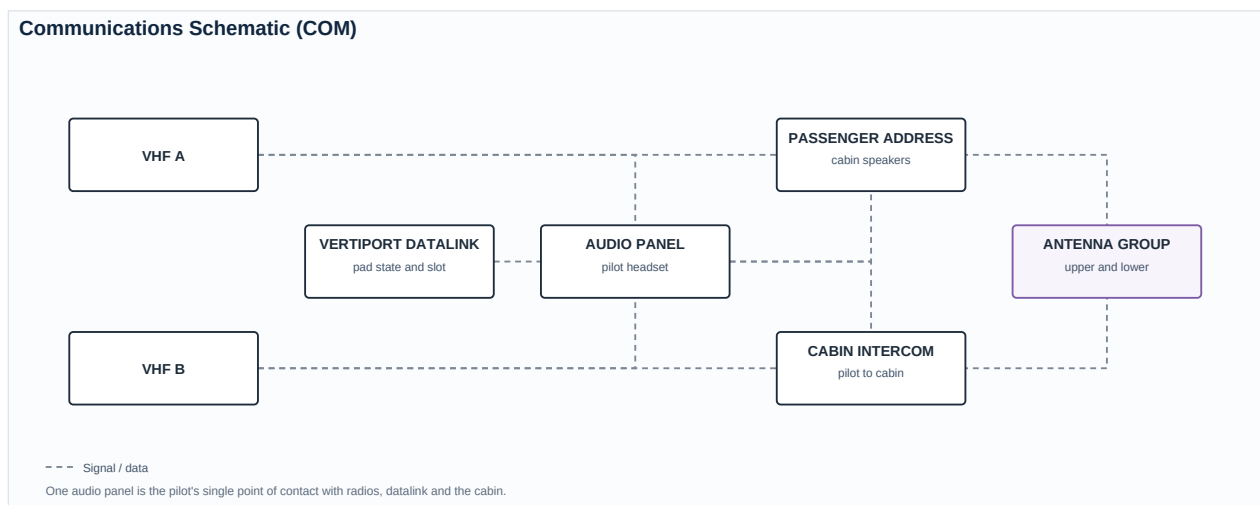


Figure 6-10 Communications (COM). Radios, datalink and cabin audio, all meeting at one audio panel.

6.10 Detect and Avoid (DAA)

Detect and Avoid Schematic (DAA)

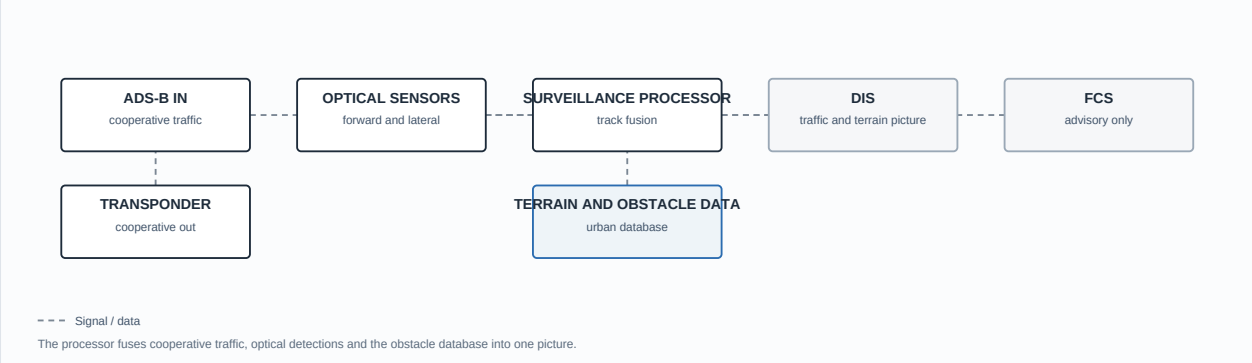


Figure 6-11 Detect and Avoid (DAA). Cooperative traffic, optical detection and the obstacle database fused into one picture.

6.11 Flight Deck Displays and Controls (DIS)

Flight Deck Displays and Controls Schematic (DIS)

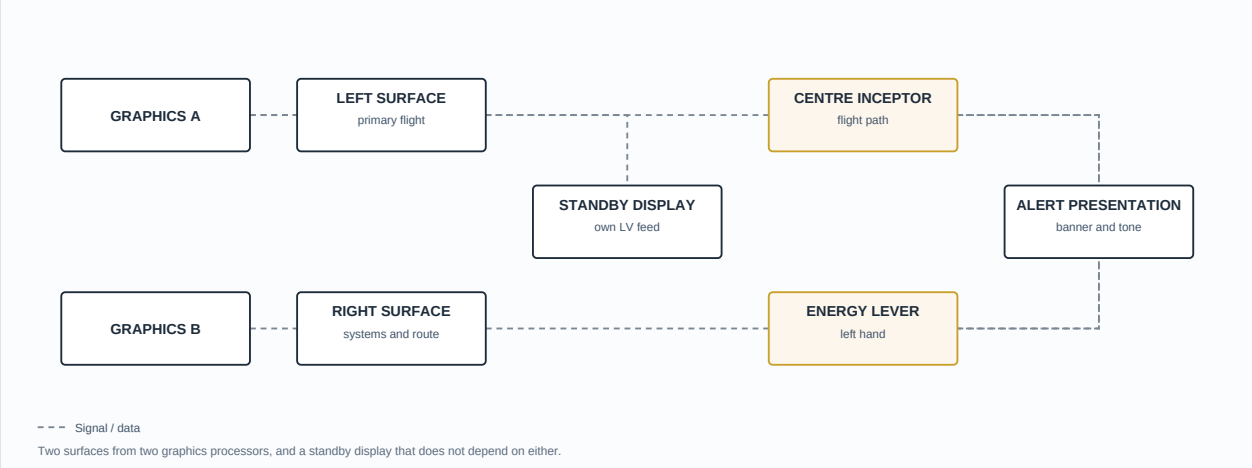


Figure 6-12 Flight Deck Displays and Controls (DIS). Two graphics processors, two surfaces, a standby display and the alert presentation path.

6.12 Landing Gear and Ground Contact (LGS)

Landing Gear and Ground Contact Schematic (LGS)

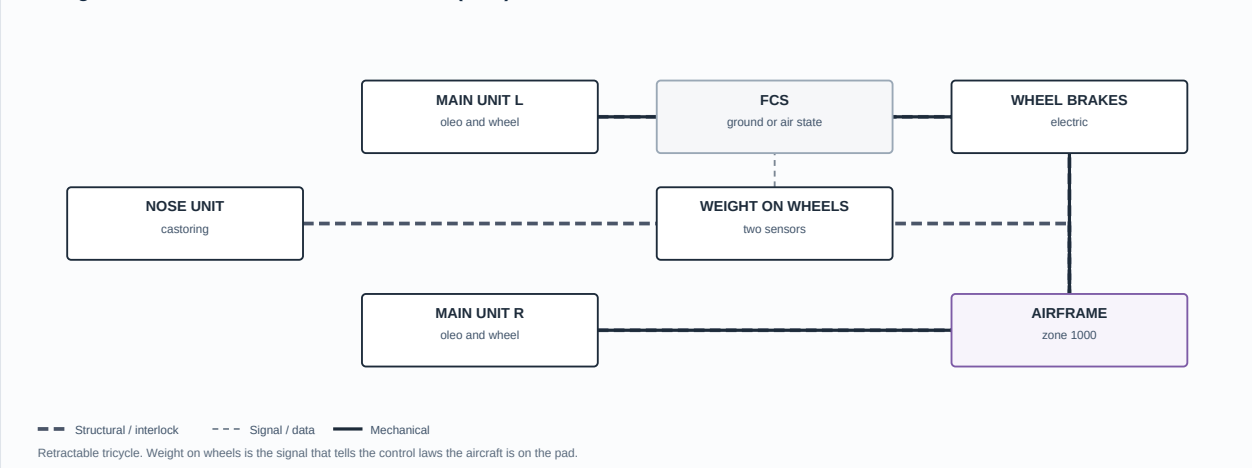


Figure 6-13 Landing Gear and Ground Contact (LGS). Gear units on the airframe, and the weight-on-wheels signal that reaches the control laws.

6.13 Airframe Structure (STR)

Airframe Structure Schematic (STR)

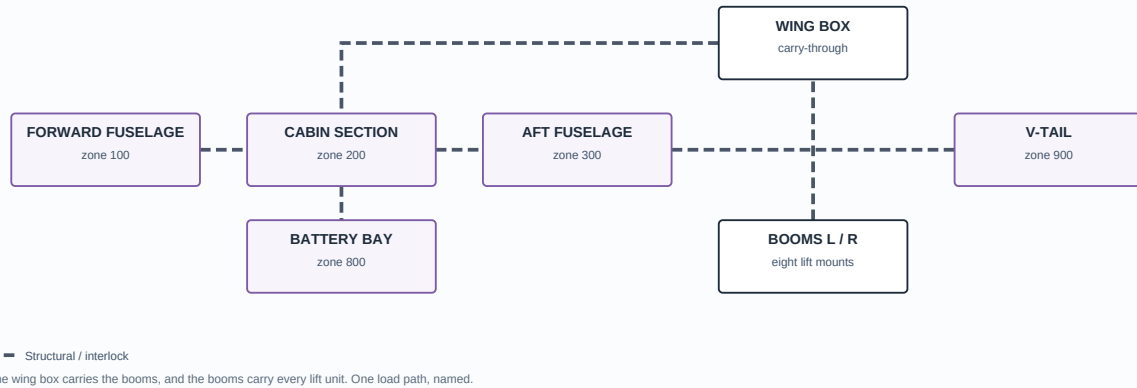


Figure 6-14 Airframe Structure (STR). The load path from the V-tail through the fuselage to the wing box and out to the booms.

6.14 Cabin and Occupant Safety (CAB)

Cabin and Occupant Safety Schematic (CAB)

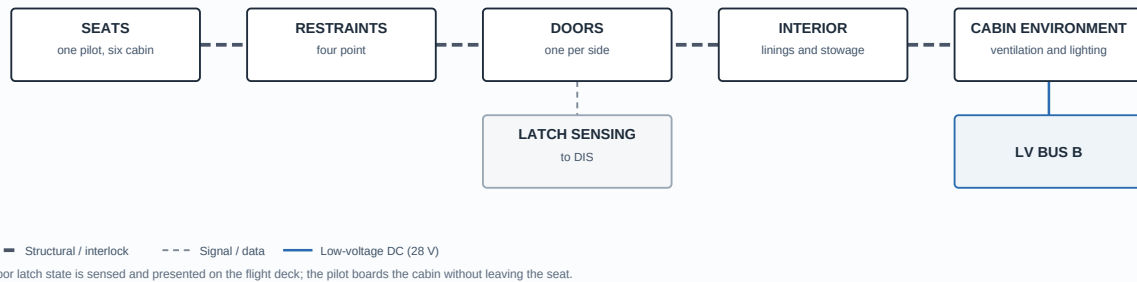


Figure 6-15 Cabin and Occupant Safety (CAB). Seats and restraints, doors and latch sensing, interior and cabin environment.

6.15 Emergency Recovery System (ERS)

Emergency Recovery System Schematic (ERS)

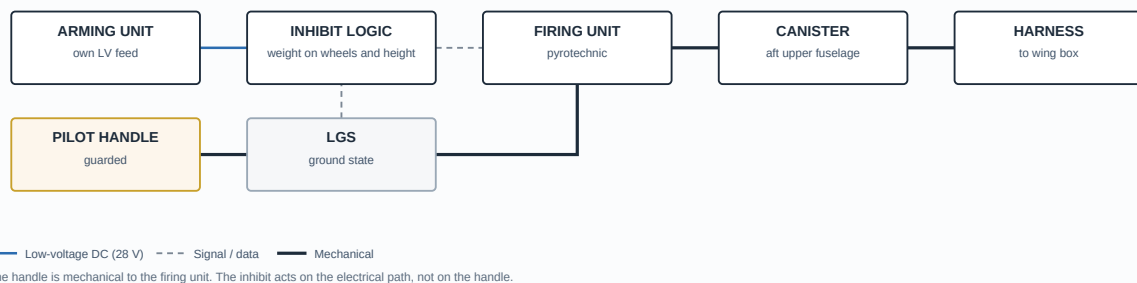


Figure 6-16 Emergency Recovery System (ERS). Arming, inhibit, firing and the mechanical path from the pilot handle to the canister.

6.16 Health Monitoring and Data Recording (HMS)

Health Monitoring and Data Recording Schematic (HMS)

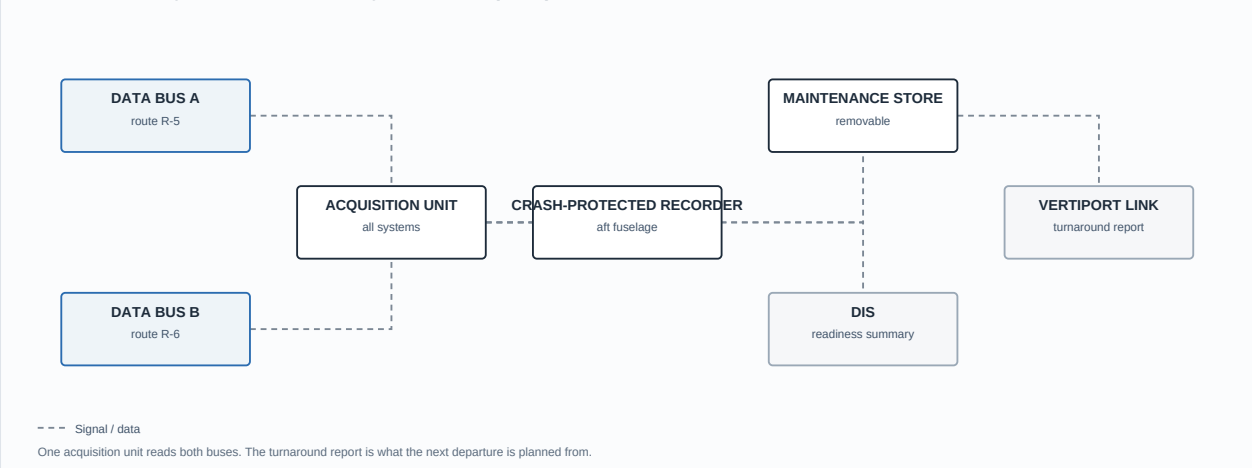


Figure 6-17 Health Monitoring and Data Recording (HMS). One acquisition unit reading both data buses, and where its output goes.

6.17 Flight Deck Arrangement

The flight deck is described as equipment in section 5.11 and as a crew station in VAY-HF-0001. The arrangement below is included so that the installation of DIS can be read without leaving this document. Panel zones, the geometry reference scheme, the control schedule and the crew procedures are in VAY-HF-0001 and are not repeated here.

Flight Deck Arrangement

Panel and seating as described in section 3. Dimensions are not to scale.

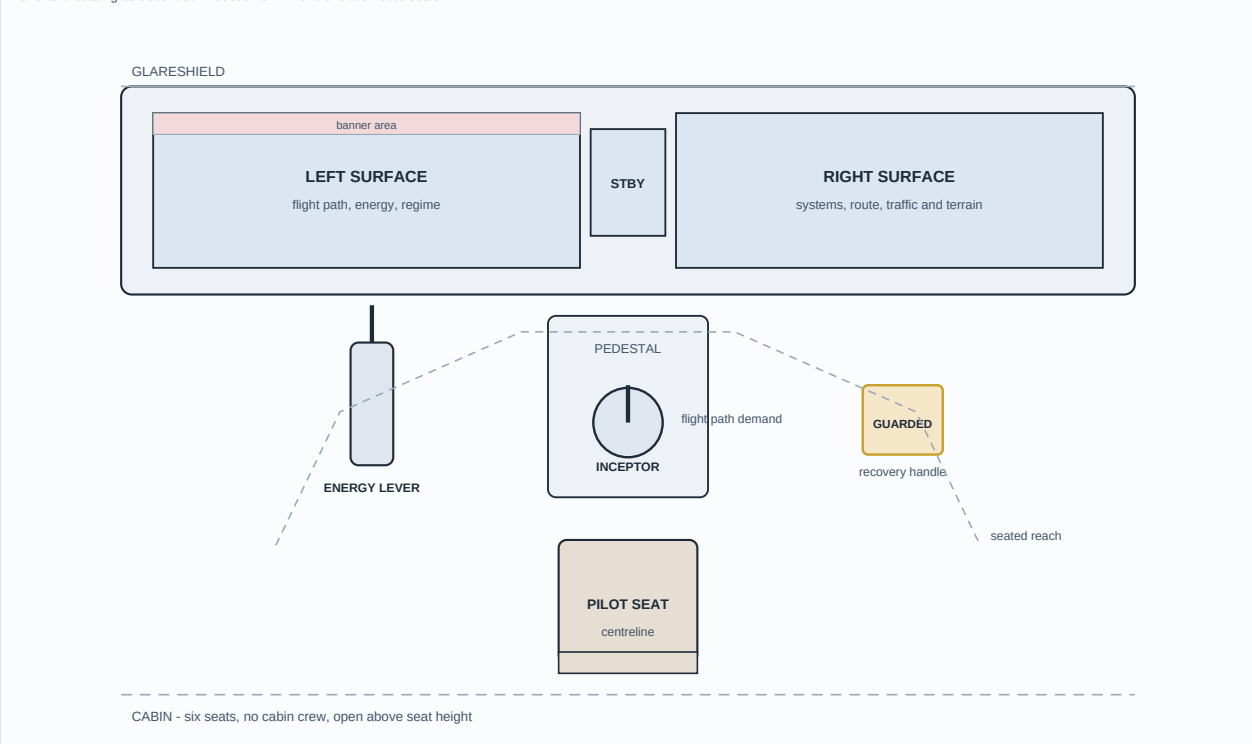


Figure 6-18 Flight deck arrangement. The panel, the inceptors and the seated reach envelope. Drawn in full on sheet HF-1 of VAY-HF-0001.

Appendix A - What this document does not do

This document is a description. It is written to be the input to a safety assessment, and it deliberately stops where the assessment begins.

It does not identify failure conditions. It says that the two booms are fed from separate high-voltage channels; it does not say what happens if a channel is lost, and it does not name that as a condition to be assessed. Section 5 describes what each system is and what it is connected to, and the work of asking what the loss or malfunction of each function would mean to the aircraft belongs to the functional hazard assessment.

It does not classify severity. The five classes are named in section 1.3 so that the reader knows the vocabulary the programme uses, and no class is applied to anything in this document.

It does not state probability objectives or development assurance levels. Those follow from the certification basis and the category, and the category is not settled here.

It contains no requirements. No sentence in it imposes an obligation on the design. Where a design property is stated, it is stated as a fact about the aircraft as it stands.

It states no safety commitment as a number. There is no crew response time, no exposure time, no demonstrated rate and no endurance figure offered as an objective. Where an assessment needs such a number it must record it as an assumption, and the assumption then belongs to whoever is able to validate it.

Read the document as a description of an aircraft that exists. Everything that follows from it - what can go wrong, how badly it would matter, how often it may be allowed to happen, and what has to be shown - is the assessment's work and not this document's.

Appendix B - Source of aircraft-level facts

The Vayu VY-6 is a demonstration article. Its aircraft-level characteristics were chosen to follow the published envelope of a real six-passenger urban eVTOL programme, so that the document describes an aircraft of a realistic size, speed, payload and mission rather than an invented one. Those characteristics, and only those, are listed below with their source.

Everything below aircraft level - the sixteen systems, their internal architecture, their redundancy structure, their interfaces, the zonal layout and every installation detail in sections 2 to 6 - is representative. It was constructed for this document and it is not the design data of any manufacturer.

Characteristic	Value used	Basis
Occupancy	1 pilot, 6 passengers	Published configuration of the Sarla Aviation Shunya, a six-passenger-plus-pilot eVTOL.
Wing span	15.00 m	Published full-scale span for the same programme.
Maximum speed	250 km/h	Published top speed for the same programme.
Payload	680 kg	Published payload for the same programme.
Typical sector	20 to 30 km	Published urban trip range for the same programme, with a longer-range growth path stated separately.
Propulsion	All-electric	Published propulsion type for the same programme.
Operating concept	Urban vertiport to vertiport, service from 2028	Published operating concept and service target for the same programme.
Certification	Small-category VTOL special condition, category not settled	Published certification intent for the same programme, with the certifying authority its own.
Length, height, mass	10.40 m, 3.60 m, 2,600 kg	Not published. Chosen to be consistent with the span, payload and occupancy above.
Propulsor arrangement	8 lift rotors, 2 cruise propellers	Not published in this form. Representative lift-plus-cruise arrangement for the class.
All system architecture	Sections 2 to 6	Not published and not manufacturer data. Constructed for this document.

Sources were public at the time of writing and were read on 2 September 2026. A programme in flight test changes; anyone relying on the aircraft-level figures should re-read the source rather than this table.